

ECE238L Computer Logic Design

Lab 7: More State Machines and Counters

Section 003: Wednesday 2:00-4:45

Name: Tomas Salaz

One Hot Counter VHDL Source Code, Simulation Code, and Constraints

```
OneHotTop.vhd
C:/Users/salaz25/ECE238L/OneHotCounter/OneHotCounter.srcs/sources_1/new/OneHotTop.vhd
9 -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19
20
21
22
23 library IEEE;
24 use IEEE.STD_LOGIC_1164.ALL;
25
26 -- Uncomment the following library declaration if using
27 -- arithmetic functions with Signed or Unsigned values
28 --use IEEE.NUMERIC_STD.ALL;
29
30 -- Uncomment the following library declaration if instantiating
31 -- any Xilinx leaf cells in this code.
32 --library UNISIM;
33 --use UNISIM.VComponents.all;
34
35 entity OneHotTop is
36   Port ( Clock_System : in STD_LOGIC;
37         Reset : in STD_LOGIC;
38         Direction : in STD_LOGIC;
39         Count : out STD_LOGIC_VECTOR (5 downto 0));
40 end OneHotTop;
41
42 architecture Behavioral of OneHotTop is
43   component FreqDivider
44     Port ( Clock_System : in STD_LOGIC;
45           Clock_IHz : out STD_LOGIC);
46   end component;
47
48   component OneHotCounter
49     Port ( Clock_System : in STD_LOGIC;
50           Direction : in STD_LOGIC;
51           Reset : in STD_LOGIC;
52           Count : out STD_LOGIC_VECTOR (5 downto 0));
53   end component;
54
55   signal slowClock : STD_LOGIC := '0';
56
57   begin
58     fd : FreqDivider port map (Clock_System => Clock_System,
59                               Clock_IHz => slowClock);
60     hcc : OneHotCounter port map (Clock_System => slowClock, Reset => Reset,
61                                  Direction => Direction, Count => Count);
62   end Behavioral;
```

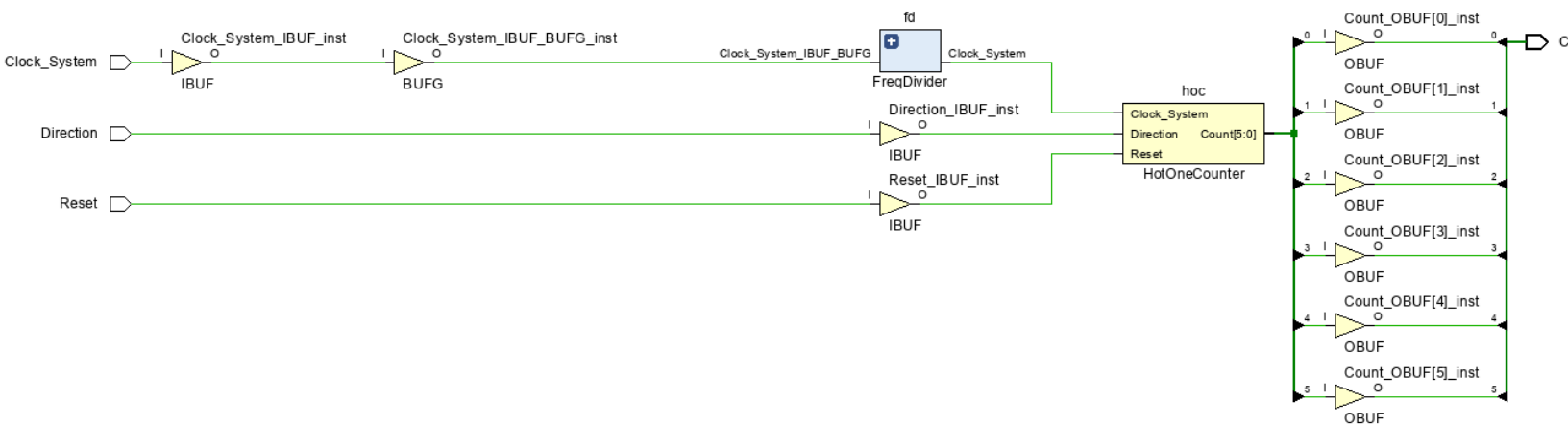
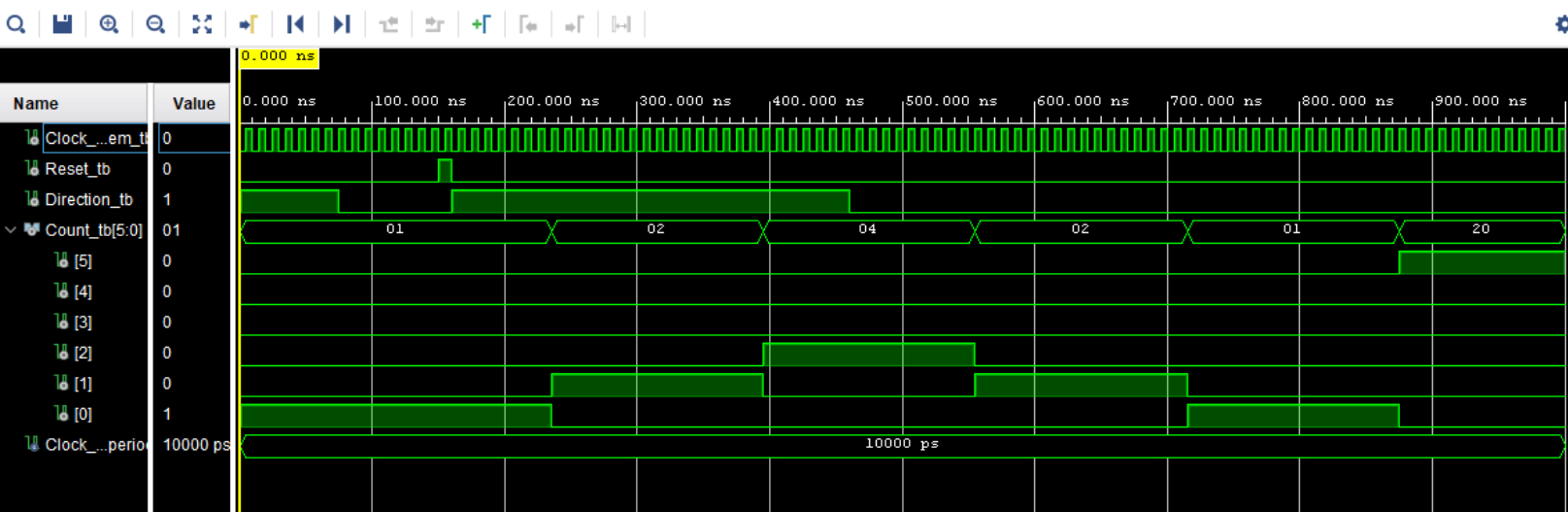
```
OneHotCounter.vhd
C:/Users/salaz25/ECE238L/OneHotCounter/OneHotCounter.srcs/sources_1/new
35 Port ( Clock_System : in STD_LOGIC;
36         Direction : in STD_LOGIC;
37         Reset : in STD_LOGIC;
38         Count : out STD_LOGIC_VECTOR (5 downto 0));
39 end OneHotCounter;
40
41 architecture Behavioral of OneHotCounter is
42
43   type state_type is (s1, s2, s4, s8, s16, s32);
44   signal presState, nextState : state_type;
45
46   begin
47     process (Clock_System, Reset)
48     begin
49       if Reset = '1' then
50         presState <= s1;
51       elsif rising_edge (Clock_System) then
52         presState <= nextState;
53       end if;
54     end process;
55
56     process (presState, Direction)
57     begin
58       case presState is
59         when s1 =>
60           Count <= "000001";
61           if Direction = '1' then nextState <= s2;
62           else nextState <= s32; end if;
63         when s2 =>
64           Count <= "000010";
65           if Direction = '1' then nextState <= s4;
66           else nextState <= s1; end if;
67         when s4 =>
68           Count <= "000100";
69           if Direction = '1' then nextState <= s8;
70           else nextState <= s2; end if;
71         when s8 =>
72           Count <= "001000";
73           if Direction = '1' then nextState <= s16;
74           else nextState <= s4; end if;
75         when s16 =>
76           Count <= "010000";
77           if Direction = '1' then nextState <= s32;
78           else nextState <= s8; end if;
79         when s32 =>
80           Count <= "100000";
81           if Direction = '1' then nextState <= s1;
82           else nextState <= s16; end if;
83       end case;
84     end process;
85   end Behavioral;
```

```
FreqDivider.vhd
C:/Users/salaz25/ECE238L/FreqDivider/FreqDivider.srcs/sources_1/new/FreqDivider.vhd
5 -- Create Date: 03/11/2025 01:26:10 PM
6 -- Design Name:
7 -- Module Name: FreqDivider - Behavioral
8 -- Project Name:
9 -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24 use IEEE.NUMERIC_STD.ALL;
25
26 -- Uncomment the following library declaration if using
27 -- arithmetic functions with Signed or Unsigned values
28 --use IEEE.NUMERIC_STD.ALL;
29
30 -- Uncomment the following library declaration if instantiating
31 -- any Xilinx leaf cells in this code.
32 --library UNISIM;
33 --use UNISIM.VComponents.all;
34
35 entity FreqDivider is
36   Port ( Clock_System : in STD_LOGIC;
37         Clock_IHz : out STD_LOGIC);
38 end FreqDivider;
39
40 architecture Behavioral of FreqDivider is
41   signal count : INTEGER := 1;
42   signal sClock_IHz : STD_LOGIC := '0';
43
44   begin
45     process (Clock_System)
46     begin
47       if Clock_System' event and Clock_System = '1' then
48         count <= count + 1;
49         if count = 50000000 then --LINE FOR CHANGING
50           Clock_IHz <= not sClock_IHz;
51           sClock_IHz <= not sClock_IHz;
52           count <= 1;
53         end if;
54       end if;
55     end process;
56   end Behavioral;
```

```
OneHotCounter_tb.vhd
C:/Users/salaz25/ECE238L/OneHotCounter/OneHotCounter.srcs/sim_1/new/OneHotCounter_tb.v
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity OneHotCounter_tb is
35   -- Port ( );
36 end OneHotCounter_tb;
37
38 architecture Bench of OneHotCounter_tb is
39   component OneHotTop
40     Port ( Clock_System : in STD_LOGIC;
41           Reset : in STD_LOGIC;
42           Direction : in STD_LOGIC;
43           Count : out STD_LOGIC_VECTOR (5 downto 0));
44   end component;
45
46   signal Clock_System_tb : STD_LOGIC;
47   signal Reset_tb : STD_LOGIC;
48   signal Direction_tb : STD_LOGIC;
49   signal Count_tb : STD_LOGIC_VECTOR(5 downto 0);
50   constant Clock_1000MHz_period : time := 10ns;
51
52   begin
53     uut : OneHotTop Port map (Clock_System => Clock_System_tb,
54                               Reset => Reset_tb,
55                               Direction => Direction_tb,
56                               Count => Count_tb);
57
58     clk_process : process
59     begin
60       Clock_System_tb <= '0'; wait for Clock_1000MHz_period/2;
61       Clock_System_tb <= '1'; wait for Clock_1000MHz_period/2;
62     end process;
63
64     stimulus : process
65     begin
66       Reset_tb <= '0';
67       Direction_tb <= '1'; wait for 75 ns;
68       Direction_tb <= '0'; wait for 75 ns;
69       Reset_tb <= '1'; wait for 10 ns;
70       Reset_tb <= '0';
71       Direction_tb <= '1'; wait for 300 ns;
72       Direction_tb <= '0'; wait for 300 ns;
73
74       wait;
75     end process;
76   end Bench;
```

```
OneHot_Constr.xdc
C:/Users/salaz25/ECE238L/OneHotCounter/OneHotCounter.srcs/constrs_1/new/OneHot_Constr
1 set_property IOSTANDARD LVCMOS33 [get_ports {Count[5]}]
2 set_property IOSTANDARD LVCMOS33 [get_ports {Count[4]}]
3 set_property IOSTANDARD LVCMOS33 [get_ports {Count[3]}]
4 set_property IOSTANDARD LVCMOS33 [get_ports {Count[2]}]
5 set_property IOSTANDARD LVCMOS33 [get_ports {Count[1]}]
6 set_property IOSTANDARD LVCMOS33 [get_ports {Count[0]}]
7 set_property IOSTANDARD LVCMOS33 [get_ports {Clock_System}]
8 set_property IOSTANDARD LVCMOS33 [get_ports {Direction}]
9 set_property IOSTANDARD LVCMOS33 [get_ports {Reset}]
10 set_property PACKAGE_PIN E3 [get_ports {Clock_System}]
11 set_property PACKAGE_PIN J15 [get_ports {Reset}]
12 set_property PACKAGE_PIN L16 [get_ports {Direction}]
13 set_property PACKAGE_PIN H17 [get_ports {Count[0]}]
14 set_property PACKAGE_PIN K15 [get_ports {Count[1]}]
15 set_property PACKAGE_PIN J13 [get_ports {Count[2]}]
16 set_property PACKAGE_PIN N14 [get_ports {Count[3]}]
17 set_property PACKAGE_PIN R18 [get_ports {Count[4]}]
18 set_property PACKAGE_PIN V17 [get_ports {Count[5]}]
19
```

One Hot Counter Waveform and Schematic



Ten State Counter VHDL Source Code, Simulation Code and Constraints

```
FreqDivider.vhd
C:/Users/salaz25/ECE238L/FreqDivider/FreqDivider.srcs/sources_1/new
5 -- Create Date: 03/11/2025 01:26:10 PM
6 -- Design Name:
7 -- Module Name: FreqDivider - Behavioral
8 -- Project Name:
9 -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24 use IEEE.NUMERIC_STD.ALL;
25
26 -- Uncomment the following library declaration if us
27 -- arithmetic functions with Signed or Unsigned valu
28 --use IEEE.NUMERIC_STD.ALL;
29
30 -- Uncomment the following library declaration if in
31 -- any Xilinx leaf cells in this code.
32 --library UNISIM;
33 --use UNISIM.VComponents.all;
34
35 entity FreqDivider is
36   Port ( Clock_System : in STD_LOGIC;
37         Clock_1Hz : out STD_LOGIC);
38 end FreqDivider;
39
40 architecture Behavioral of FreqDivider is
41   signal count : INTEGER := 1;
42   signal sClock_1Hz : STD_LOGIC := '0';
43
44   begin
45     process (Clock_System)
46     begin
47       if Clock_System' event and Clock_System = '1' then
48         count <= count + 1;
49         if count = 50000000 then --LIN
50           Clock_1Hz <= not sClock_1Hz;
51           sClock_1Hz <= not sClock_1Hz;
52           count <= 1;
53         end if;
54       end if;
55     end process;
56 end Behavioral;
```

```
TenStateCounter.vhd
C:/Users/salaz25/ECE238L/TenStateCounter/TenStateCounter.srcs/sources_1/new
28 -- arithmetic functions with Signed or Unsigned values
29 --use IEEE.NUMERIC_STD.ALL;
30
31 -- Uncomment the following library declaration if instantiat
32 -- any Xilinx leaf cells in this code.
33 --library UNISIM;
34 --use UNISIM.VComponents.all;
35
36 entity TenStateCounter is
37   Port ( Clock_System : in STD_LOGIC;
38         Direction : in STD_LOGIC;
39         Reset : in STD_LOGIC;
40         Count : out STD_LOGIC_VECTOR (3 downto 0));
41 end TenStateCounter;
42
43 architecture Behavioral of TenStateCounter is
44   type state_type is (s0, s1, s2, s3, s4, s5, s6, s7, s8, s9);
45   signal presentState, nextState : state_type;
46
47   begin
48     syncProc : process (Clock_System, Reset)
49     begin
50       if Reset = '1' then
51         presentState <= s0;
52       elsif rising_edge (Clock_System) then
53         presentState <= nextState;
54       end if;
55     end process;
56
57     comboProc : process (presentState, Direction)
58     begin
59       case presentState is
60         when s0 => Count <= "0000";
61         if Direction = '1' then nextState <= s1;
62         else nextState <= s9;
63         end if;
64         when s1 => Count <= "0001"; nextState <= s2;
65         when s2 => Count <= "0010"; nextState <= s3;
66         when s3 => Count <= "0011"; nextState <= s4;
67         when s4 => Count <= "0100"; nextState <= s0;
68         when s5 => Count <= "0101"; nextState <= s9;
69         when s6 => Count <= "0110"; nextState <= s5;
70         when s7 => Count <= "0111"; nextState <= s6;
71         when s8 => Count <= "1000"; nextState <= s7;
72         when s9 => Count <= "1001";
73         if Direction = '0' then nextState <= s8;
74         else nextState <= s0;
75         end if;
76       end case;
77     end process;
78 end Behavioral;
```

```
DisplayDigits.vhd
C:/Users/salaz25/ECE238L/DisplayDigits/DisplayDigits.srcs/sources_1/new
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiat
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity DisplayDigits is
35   Port ( Count : in STD_LOGIC_VECTOR (3 downto 0);
36         Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0)
37         Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
38 end DisplayDigits;
39
40 architecture Behavioral of DisplayDigits is
41   begin
42     process (Count)
43     begin
44       if Count = "0000" then Cathode_7SD <= "00000011"; --0
45       elsif Count = "0010" then Cathode_7SD <= "10011111";
46       elsif Count = "0011" then Cathode_7SD <= "00100101";
47       elsif Count = "0100" then Cathode_7SD <= "00001101";
48       elsif Count = "0101" then Cathode_7SD <= "10011001";
49       elsif Count = "0110" then Cathode_7SD <= "11000001";
50       elsif Count = "0111" then Cathode_7SD <= "00011111";
51       elsif Count = "1000" then Cathode_7SD <= "00000001";
52       elsif Count = "1001" then Cathode_7SD <= "00001001";
53
54       elsif Count = "1010" then Cathode_7SD <= "00010001";
55       elsif Count = "1011" then Cathode_7SD <= "11000001";
56       elsif Count = "1100" then Cathode_7SD <= "01100011";
57       elsif Count = "1101" then Cathode_7SD <= "10000101";
58       elsif Count = "1110" then Cathode_7SD <= "01100001";
59       elsif Count = "1111" then Cathode_7SD <= "01110001";
60     end if;
61     Anode_7SD <= "11111110";
62   end process;
63 end Behavioral;
```

```
TenStateTop.vhd
C:/Users/salaz25/ECE238L/TenStateCounter/TenStateCounter.srcs/sources_1/new
32 --use UNISIM.VComponents.all;
33
34 entity TenStateTop is
35   Port (
36     Clock_System : in STD_LOGIC;
37     Direction : in STD_LOGIC;
38     Reset : in STD_LOGIC;
39     Cathode7SD : out STD_LOGIC_VECTOR(7 downto 0);
40     Anode7SD : out STD_LOGIC_VECTOR(7 downto 0);
41   );
42 end TenStateTop;
43
44 architecture Behavioral of TenStateTop is
45   component FreqDivider
46   Port (
47     Clock_System : in STD_LOGIC;
48     Clock_1Hz : out STD_LOGIC);
49   end component;
50   component DisplayDigits
51   Port (
52     Count : in STD_LOGIC_VECTOR(3 downto 0);
53     Cathode_7SD : out STD_LOGIC_VECTOR(7 downto 0);
54     Anode_7SD : out STD_LOGIC_VECTOR(7 downto 0));
55   end component;
56   component TenStateCounter
57   Port (
58     Clock_System : in STD_LOGIC;
59     Direction : in STD_LOGIC;
60     Reset : in STD_LOGIC;
61     Count : out STD_LOGIC_VECTOR(3 downto 0));
62   end component;
63
64   signal slowClock : STD_LOGIC := '0';
65   signal Count : STD_LOGIC_VECTOR(3 downto 0) := "0000";
66
67   begin
68     FD : FreqDivider
69     port map (
70       Clock_System => Clock_System,
71       Clock_1Hz => slowClock);
72
73     TSC : TenStateCounter port map (
74       Clock_System => slowClock,
75       Reset => Reset,
76       Direction => Direction,
77       Count => Count);
78
79     DD : DisplayDigits port map (
80       Count => Count,
81       Cathode_7SD => Cathode7SD,
82       Anode_7SD => Anode7SD );
83 end Behavioral;
```

Ten State Counter Code Continued

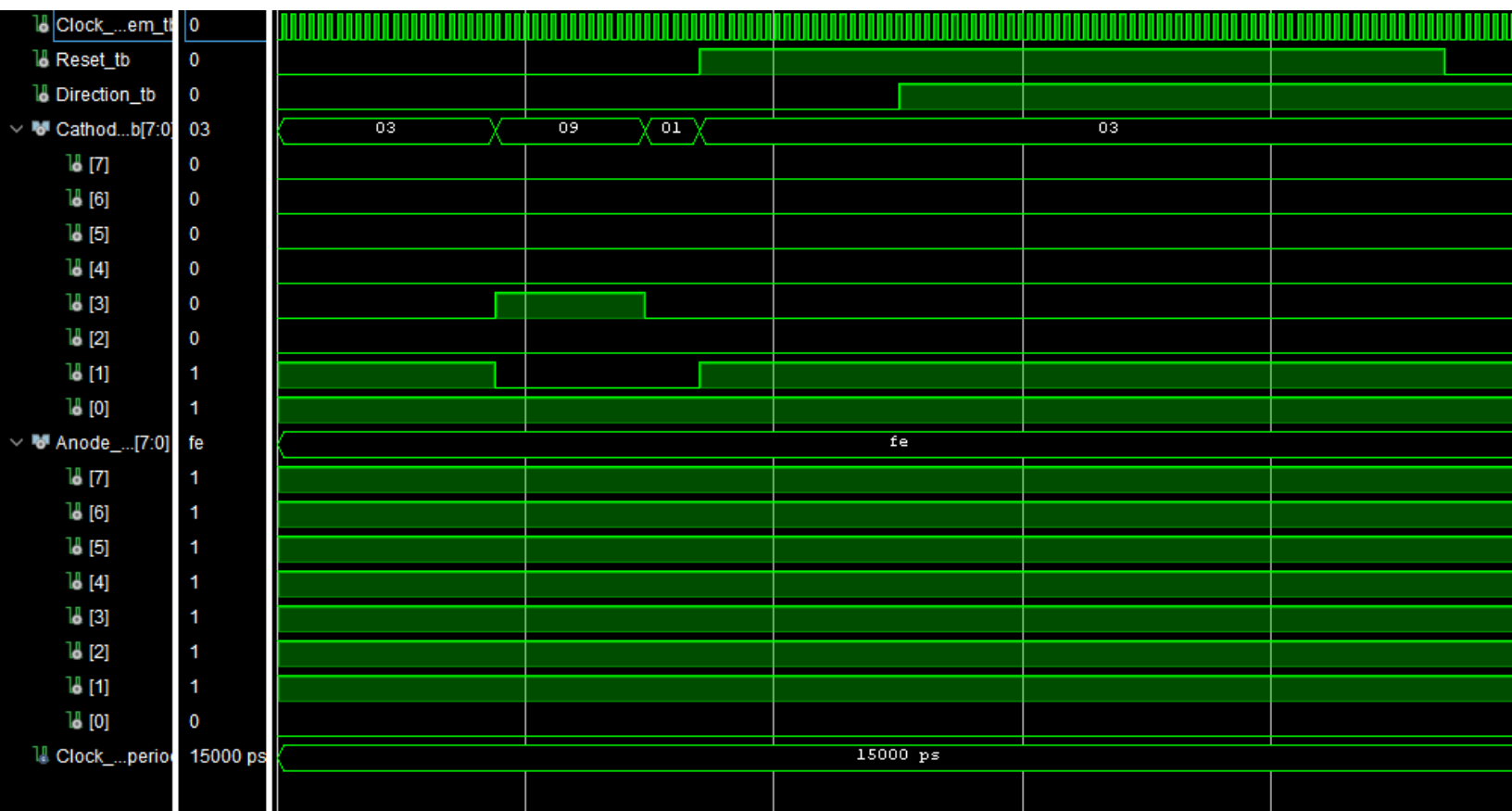
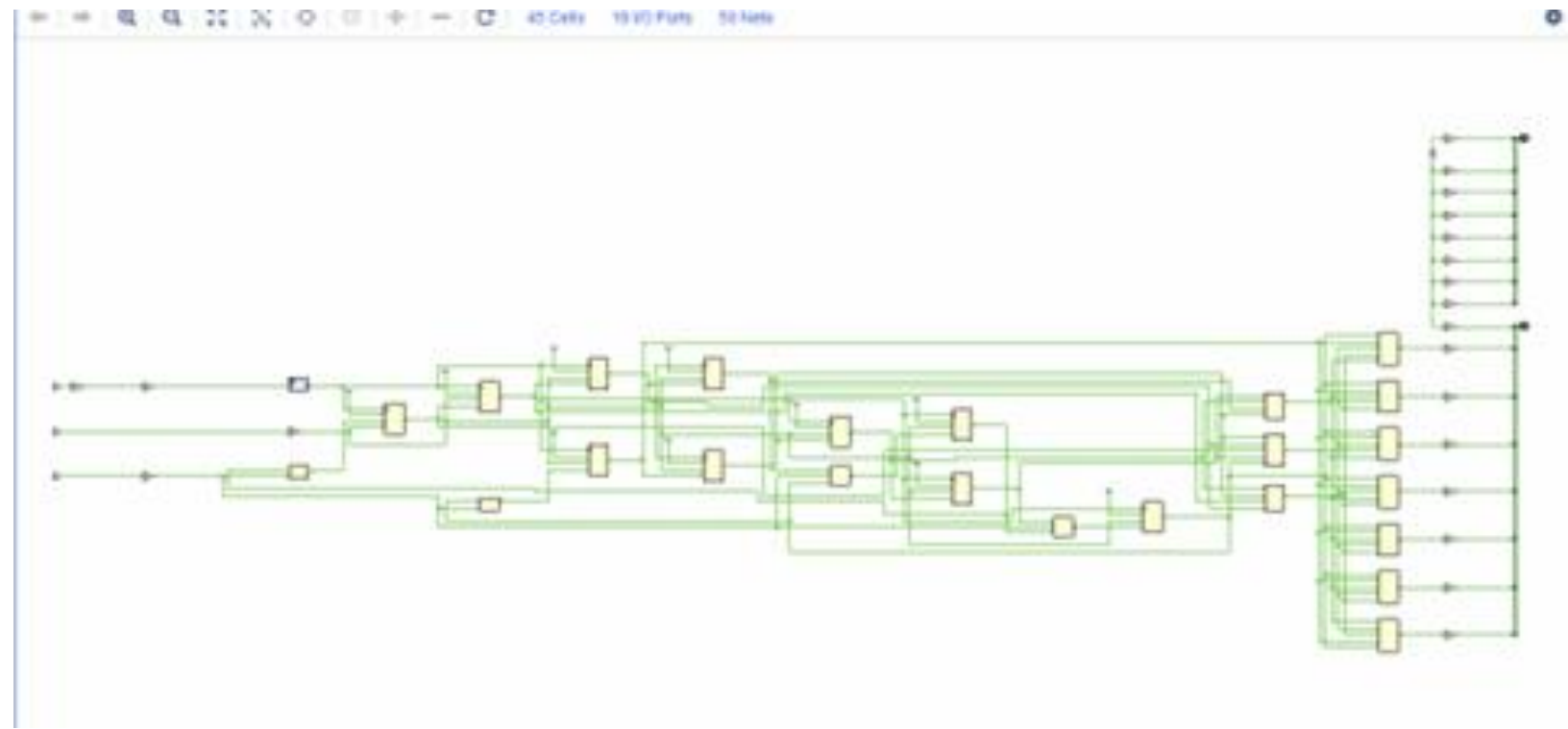
```
TenState_tb.vhd
C:/Users/salaz25/ECE238L/TenStateCounter/TenStateCounter.srcs/sim_1/new/TenState_tb.vhd

28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity TenState_tb is
35 -- Port ( );
36 end TenState_tb;
37
38 architecture Behavioral of TenState_tb is
39     component TenStateTop is
40     Port (
41         Clock_System : in STD_LOGIC;
42         Direction     : in STD_LOGIC;
43         Reset         : in STD_LOGIC;
44         Cathode7SD    : out STD_LOGIC_VECTOR(7 downto 0);
45         Anode7SD      : out STD_LOGIC_VECTOR(7 downto 0);
46     end component;
47
48     signal Clock_System_tb, Reset_tb, Direction_tb : STD_LOGIC;
49     signal Cathode7SD_tb, Anode_7SD_tb : STD_LOGIC_VECTOR (7 downto 0);
50     constant Clock_1000MHz_period : time := 15ns;
51
52     begin
53     uut : TenStateTop port map (
54         Clock_System => Clock_System_tb,
55         Direction     => Direction_tb,
56         Reset         => Reset_tb,
57         Cathode7SD    => Cathode7SD_tb,
58         Anode7SD      => Anode_7SD_tb );
59
60     clk_process : process
61     begin
62         Clock_System_tb <= '0'; wait for Clock_1000MHz_period/4;
63         Clock_System_tb <= '1'; wait for Clock_1000MHz_period/4;
64     end process;
65
66     direction : process
67     begin
68         Direction_tb <= '0'; wait for 500 ns;
69         Direction_tb <= '1'; wait for 500 ns;
70     end process;
71
72     stimulus : process
73     begin
74         Reset_tb <= '0'; wait for 300ns;
75         Reset_tb <= '0'; wait for 40ns;
76         Reset_tb <= '1'; wait for 600ns;
77     end process;
78
79 end Behavioral;
80
```

```
TenState_const.xdc
C:/Users/salaz25/ECE238L/TenStateCounter/TenStateCounter.srcs/constrs_1/nc

1 set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[7]}]
2 set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[6]}]
3 set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[5]}]
4 set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[4]}]
5 set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[3]}]
6 set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[2]}]
7 set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[1]}]
8 set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[0]}]
9 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[7]}]
10 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[6]}]
11 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[5]}]
12 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[4]}]
13 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[3]}]
14 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[2]}]
15 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[1]}]
16 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[0]}]
17 set_property IOSTANDARD LVCMOS33 [get_ports Clock_System]
18 set_property IOSTANDARD LVCMOS33 [get_ports Direction]
19 set_property IOSTANDARD LVCMOS33 [get_ports Reset]
20 set_property PACKAGE_PIN E3 [get_ports Clock_System]
21 set_property PACKAGE_PIN L16 [get_ports Direction]
22 set_property PACKAGE_PIN J15 [get_ports Reset]
23 set_property PACKAGE_PIN J17 [get_ports {Anode7SD[0]}]
24 set_property PACKAGE_PIN J18 [get_ports {Anode7SD[1]}]
25 set_property PACKAGE_PIN T9 [get_ports {Anode7SD[2]}]
26 set_property PACKAGE_PIN J14 [get_ports {Anode7SD[3]}]
27 set_property PACKAGE_PIN P14 [get_ports {Anode7SD[4]}]
28 set_property PACKAGE_PIN T14 [get_ports {Anode7SD[5]}]
29 set_property PACKAGE_PIN K2 [get_ports {Anode7SD[6]}]
30 set_property PACKAGE_PIN U13 [get_ports {Anode7SD[7]}]
31 set_property PACKAGE_PIN H15 [get_ports {Cathode7SD[0]}]
32 set_property PACKAGE_PIN L18 [get_ports {Cathode7SD[1]}]
33 set_property PACKAGE_PIN T11 [get_ports {Cathode7SD[2]}]
34 set_property PACKAGE_PIN P15 [get_ports {Cathode7SD[3]}]
35 set_property PACKAGE_PIN K13 [get_ports {Cathode7SD[4]}]
36 set_property PACKAGE_PIN K16 [get_ports {Cathode7SD[5]}]
37 set_property PACKAGE_PIN R10 [get_ports {Cathode7SD[6]}]
38 set_property PACKAGE_PIN T10 [get_ports {Cathode7SD[7]}]
39
```

Ten State Counter Waveform and Schematic



Summary:

Both parts of this lab build on what we have learned since lab 5. The goal of this lab was to make more state machine counters and display the counters in more complex ways, we also added a Direction to the counting which made this lab a bit more challenging. The One Hot Counter behaved in the same way as the Grey Code Counter except that we had to implement the bit more code for the Direction functionality. The process was similar, as I implemented a Top Module with a Frequency Divider Component and a One Hot Counter Component to have the Display, the Divider and the Counting all separate.

Part B was similar, as we had to implement the same kind of counter, the differences being that this Ten State Counter was displayed on the 7-segment Display and, depending on direction, counted from 0-4 or down from 9-5. I implemented the same structure in this lab, having a Top Module for the Display, A Frequency Divider and a Counter, with the same Structure. The only small difference is that the Top module had "Display Digits" (lab 5c) to help with the 7 Segment Display.

Problems:

This lab took me longer than it should have, I kept making syntax mistakes and mixing up the name of the components. Other than there, there weren't any technical problems as the lab was relatively the same as the last lab.

Hints:

Attention to Detail matters.

Recommendations:

None, It was a good lab.